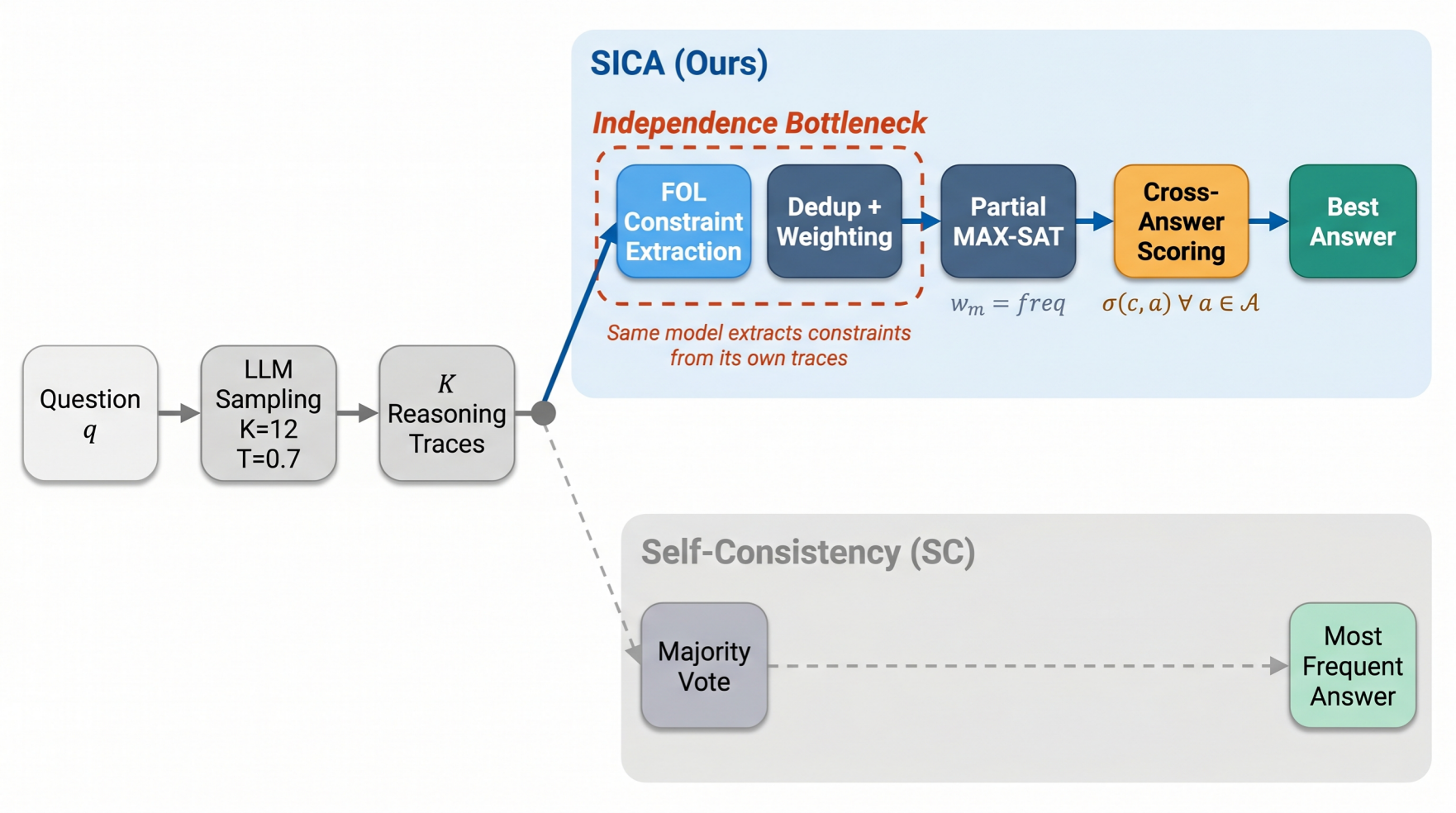

SICA (Ours)
Independence Bottleneck
FOL Constraint Extraction
Dedup + Weighting
Partial MAX-SAT
Cross-Answer Scoring
Best Answer
$w_m = freq$
$\sigma(c, a)\ \forall\ a \in \mathcal{A}$
Same model extracts constraints from its own traces
Question q
LLM Sampling K=12 T=0.7
K Reasoning Traces
Self-Consistency (SC)
Majority Vote
Most Frequent Answer